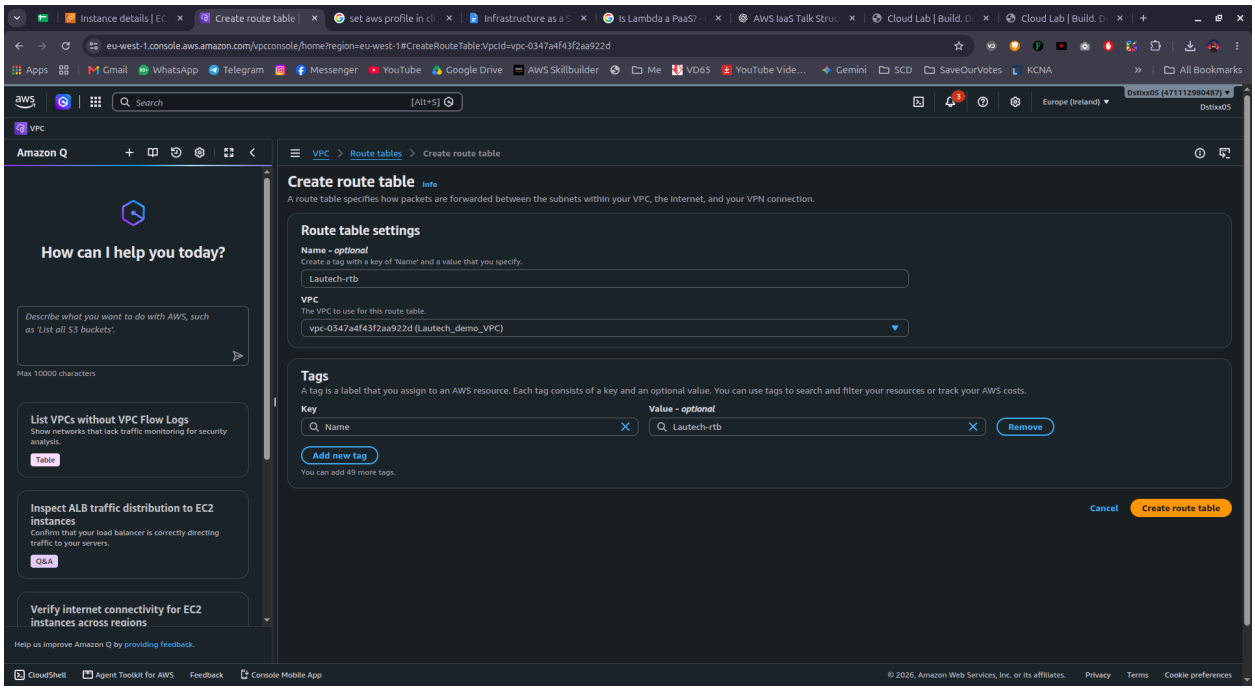
1. Navigate to **cloud-lab-react-source/cloud-lab** where you can see the package.json file.
2. Run npm install to install the dependencies. Meanwhile, you must have node installed on your PC.
3. **npm run dev** to run the react server on your machine and **npm run build** to build the prod ready files.

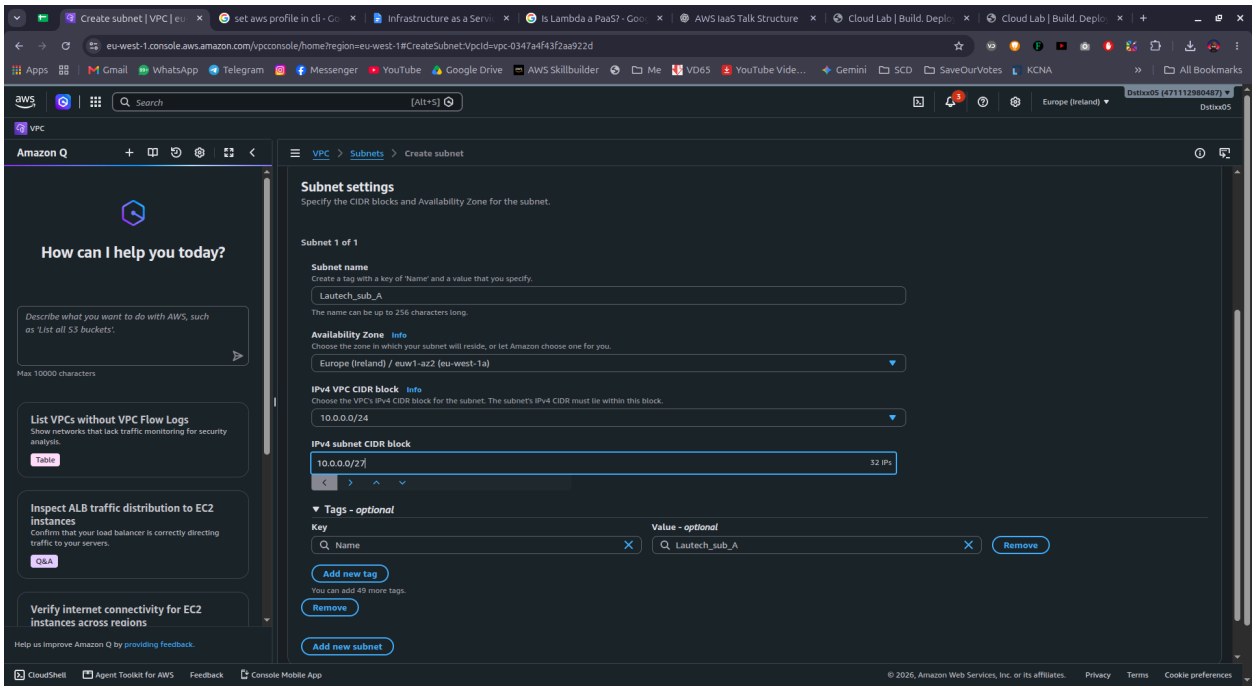
On AWS management console

1. Create a VPC. CIDR block 10.0.0.0/24
2. Create an attach Internet gateway to your newly created VPC
3. Create route table to route traffic to the internet gateway.

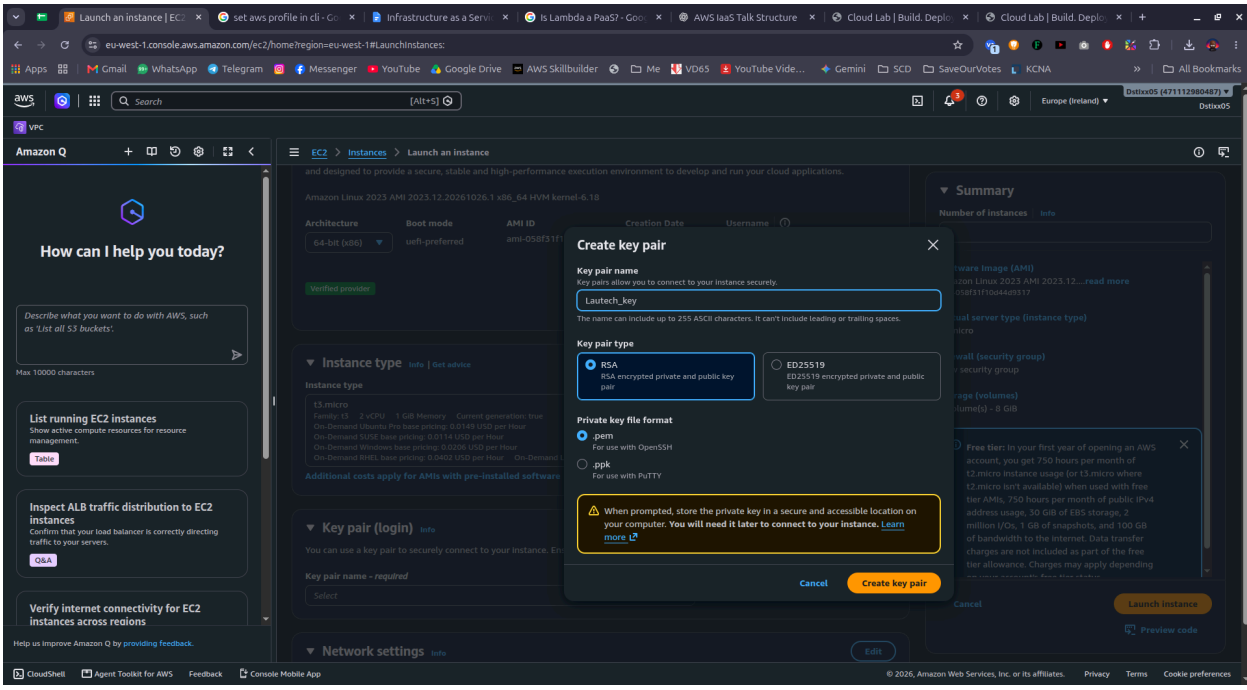


4. Add route to send traffic to IGW

5. Pick an AZ. Very important for our next hands-on: Europe (Ireland) / euw1-az2 (eu-west-1a)



6.
7. Time to set up our EC2
    a. Go to the EC2 console
    b. Create a key pair and save it on your machine. It'll be used to SSH into the instance later on.



    c. Paste the Userdata

```
#!/bin/bash
set -euxo pipefail

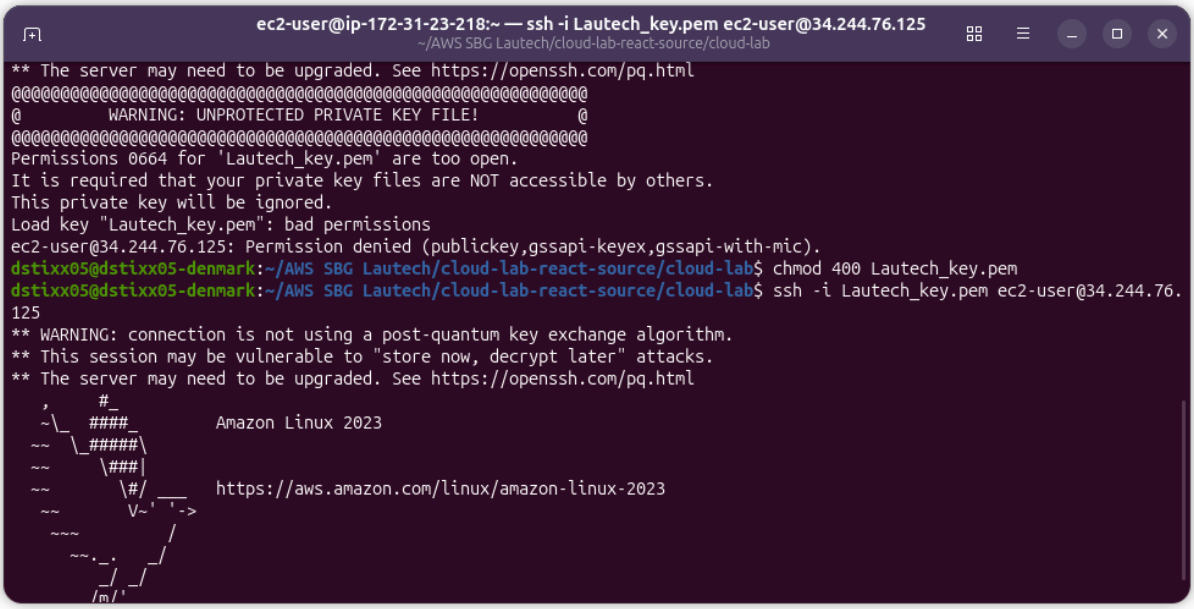
dnf update -y
dnf install -y nginx

mkdir -p /var/www/html
chown -R ec2-user:ec2-user /var/www/html

systemctl enable nginx
systemctl start nginx
```

Don’t also forget to set your default ec2 sg to anywhere ipv4.

8. Set permission to allow only owner to be able to the read the pem file



You should be able to SSH into the server.

9. We’re copying the static files to our ec2 instance so it doesn’t show this stale NGINX page when we hit the IP address

scp -r -i Lautech_key.pem ./dist ec2-user@3.252.100.249:/home/ec2-user

Copy files to **sudo cp -r /home/ec2-user/dist/* /var/www/html/**

Also create a new nginx config file at /etc/nginx/conf.d/lautech.conf

```
server {
    # Listen on port 80 (HTTP) for both IPv4 and IPv6
    listen 80;
    listen [::]:80;
```

```
    # Replace with your EC2 Public IP or your domain name
    server_name 3.252.100.249;

    # The absolute path to the directory containing your HTML files
    root /var/www/html/dist;

    # The default file Nginx looks for when a user visits the root URL
    index index.html index.htm;

    location / {
        # Check if the requested file exists, then folder, otherwise throw a 404
        try_files $uri $uri/ =404;
    }

    # Optional: Cache control for static assets (images, CSS, JS)
    location ~* \.(css|js|jpg|jpeg|png|gif|ico|svg|woff|woff2)$ {
        expires 7d;
        add_header Cache-Control "public, no-transform";
    }

    # Configure custom logs for this specific site
    access_log /var/log/nginx/static_site_access.log;
    error_log /var/log/nginx/static_site_error.log;
}
```

For replacing IP addresses after terminating EC2 instance
sudo sed -i 's/former_ip/new_ip/g' lautech.conf

Using Amplify
cd dist
zip -r ../cloud-lab-amplify.zip .
cd ..